

Beyond Aggregate Accuracy: A Category-Balanced Benchmark of Base Language Models on Medical Reasoning under Controlled Decoding Temperature

Working draft — Paper 1 (MedQA2026). Bracketed [TBD-EXP#] items await new experiments.

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Title (working; alternates below)

Beyond Aggregate Accuracy: A Category-Balanced Benchmark of Base Language Models on Medical Reasoning under Controlled Decoding Temperature

Alternates: 1. *What Aggregate MedQA Accuracy Hides: Category-Balanced Evaluation of Base Language Models.* 2. *Temperature, Stability, and Scale: An Empirical Characterization of Base LLM Medical Reasoning.* 3. *Five Kinds of Wrong: A Reasoning-Category Benchmark for Base Medical Language Models.* 4. *Base, Not Tuned: Controlled-Decoding Evaluation of Open Language Models on USMLE-Style Questions.* 5. *Stable When Right: Decoding Stability as a Confidence Signal in Base Medical LLMs.*

Abstract

Large language models are increasingly evaluated on medical question-answering benchmarks such as MedQA, yet performance is almost always reported as a single aggregate accuracy that obscures *where* and *how reliably* a model reasons. We present a controlled empirical benchmark of five open **base** (non-instruction-tuned) language models — Qwen2.5 (7B, 14B, 32B) and OLMo-3 (7B, 32B) — on a category-balanced subset of 1,030 USMLE-style questions (206 each across Clinical Findings, Diagnosis, Mechanism/Pathophysiology, Next Step/Workup, and Treatment/Management), evaluated at three decoding temperatures (0.0, 0.3, 0.7) with three independent runs per condition. We characterize behavior along four axes: category-resolved accuracy, decoding-temperature sensitivity, run-to-run answer stability, and cross-model error structure, with appropriate uncertainty quantification (Wilson intervals, exact McNemar tests, paired bootstrap, and Holm correction). We find that aggregate accuracy systematically masks a stable category ordering — Clinical Findings is the hardest category for every model — that decoding temperature *monotonically* degrades accuracy (greedy is best) while inflating run-to-run answer instability to as much as 45–68% of questions, and that scale gains are uneven: Qwen2.5 7B significantly outperforms the four-times-larger OLMo-3 32B, while the Qwen2.5 7B→14B step is not statistically significant on this balanced set. We further show that cross-run answer stability is a strong, logit-free predictor of correctness (a +23 to +42 pp accuracy gap between stable and unstable items), a property corroborated by token-logprob calibration: greedy confidence correlates negatively with run-to-run instability ($r = -0.34$ to -0.62) and models are reasonably calibrated (ECE 0.02–0.08). These results argue that medical-reasoning evaluation should move beyond aggregate accuracy toward category-resolved, stability-aware reporting, and they motivate the verification and neuro-symbolic methods we pursue in subsequent work. We release all data, configurations, and analysis code for full reproducibility. We make no claims of clinical utility.

1. Introduction

Purpose: motivate category-balanced, stability-aware evaluation of base LLMs and state contributions.

P1 — Opening problem. Language models now answer USMLE-style medical questions at scores that exceed nominal human passing thresholds, and such results are frequently cited as evidence of “medical reasoning.” But a single accuracy number on a heterogeneous benchmark tells us little about which kinds of clinical reasoning a model has actually acquired, or how stable that performance is under ordinary decoding randomness.

P2 — Why MedQA matters. MedQA (Jin et al., 2021) is the de facto standard for medical QA: it is large, exam-derived, and multiple-choice, which makes it scalable to evaluate. Its ubiquity makes the way we report MedQA results consequential for the field’s understanding of model capability.

P3 — Why aggregate accuracy is insufficient. MedQA mixes distinct reasoning

demands — recognizing clinical findings, naming a diagnosis, explaining a mechanism, choosing the next step, selecting management — in unknown proportions. An aggregate score lets strength in easy categories compensate for weakness in hard ones, hiding systematic failure modes. We show this directly: every model we test is meaningfully worse at Clinical Findings than at Treatment/Management, a gap invisible in the aggregate.

P4 — Why category-balanced analysis matters. Evaluating on a set deliberately balanced across reasoning categories (206 questions each) removes the confound of unknown category mix and lets us compare *profiles*, not just totals. It also reveals that conclusions about scaling can flip with benchmark construction (Section 6).

P5 — Why temperature and stochasticity matter. Deployed systems sample at non-zero temperature, and reproducibility requires knowing how much answers move across runs. We treat decoding temperature as a first-class experimental variable and quantify both its effect on accuracy and on run-to-run answer instability — the latter a rarely reported but decision-relevant property.

P6 — What this paper contributes. We provide a controlled, reproducible benchmark of five open base models across a category-balanced set and three temperatures with repeated runs, plus a statistical treatment (intervals, paired tests, multiple-comparison control) that supports defensible claims about scale, temperature, and category structure.

P7 — Summary of findings. (i) A stable category-difficulty ordering hidden by aggregate accuracy; (ii) monotone accuracy degradation with temperature and steep growth in answer instability; (iii) uneven scaling, including a non-significant Qwen2.5 7B→14B step and a smaller model beating a 4× larger one across families; (iv) answer stability as a strong logit-free correctness signal.

P8 — Contributions list (see Section 1.1).

1.1 Contributions

1. A **category-balanced MedQA benchmark** (1,030 questions, 206 per reasoning category) and a reproducible evaluation harness for base LLMs across temperatures and repeated runs.
 2. A **category-resolved empirical characterization** showing that aggregate accuracy hides a consistent difficulty ordering, with uncertainty-quantified per-category results.
 3. A **decoding-temperature and stability analysis** establishing monotone degradation and quantifying run-to-run answer instability (up to 45-68% of items at $t=0.7$).
 4. A **statistically grounded scale comparison** (Wilson CIs, exact McNemar, bootstrap, Holm) yielding defensible positive and *null* results about model scale and family.
 5. Evidence that **cross-run stability predicts correctness**, motivating verification-based methods, released with full data, configs, and analysis code.
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2. Related Work

Purpose: position the paper across six clusters and name the gap.

Medical QA benchmarks. MedQA/USMLE (Jin et al., 2021), MedMCQA, PubMedQA, MMLU-medical. Cite for dataset lineage and the dominance of aggregate-accuracy reporting. *Gap:* these are typically scored in aggregate; category-resolved reasoning analysis is rare.

LLMs for clinical reasoning. Med-PaLM / Med-PaLM 2, GPT-4 medical evaluations, open medical models. *Position:* we deliberately study *base* open models to characterize raw language-model behavior, not deployed assistants, and we do not claim clinical utility.

Base vs instruction-tuned reasoning. Work on how instruction tuning and RLHF reshape reasoning and calibration. *Position:* we isolate base models to remove tuning confounds and [TBD-EXP2: add a controlled base-vs-instruct contrast].

Temperature, sampling, and stochastic decoding. Self-consistency (Wang et al.), temperature/nucleus sampling studies. *Position:* we measure temperature’s effect on both accuracy and answer instability on a fixed medical benchmark with repeated runs.

Evaluation beyond aggregate accuracy. Subgroup/slice evaluation, behavioral testing, robustness suites. *Position:* we instantiate slice-based evaluation along *reasoning categories* for medical QA and connect it to scaling conclusions.

Reliability, uncertainty, and calibration in medical AI. Calibration/ECE, selective prediction, abstention. *Position:* we show stability is a usable confidence proxy and ground it in logprob calibration (ECE + confidence–stability correlation, EXP1). *Gap this paper fills:* a controlled, category-balanced, temperature-and-stability characterization of open base models with proper statistics — a foundation absent from prior aggregate-accuracy reports.

3. Dataset

Purpose: present the balanced 1,030-question set and its construction.

The benchmark is a balanced subset of MedQA-USMLE. **MedQA itself provides no reasoning-type labels** (each item is only a question, four/five options, and an answer); the five reasoning categories are introduced in this work. We define them to map onto distinct clinical cognitive skills — **Clinical Findings, Diagnosis, Mechanism/Pathophysiology, Next Step/Workup, Treatment/Management** — and sample **206 questions per category** for a total of **1,030**, in the 4-option (A-D) format (random baseline 25%).

Category-assignment procedure. Each candidate question is labeled by **two independent methods**: (i) a rule-based matcher keyed on question phrasing (e.g., “most likely diagnosis” → Diagnosis; “next step” → Next Step/Workup; “mechanism/action” → Mechanism), and (ii) an independent LLM classifier (Qwen 2.5) that reads the full

question. **We retain only the questions on which the two methods agree** and discard disagreements as ambiguous. On the validated pool this dual-method agreement was 68.4% (1,206 / 1,764 kept; 558 discarded), and agreement was lowest for Treatment/Management (54%), which becomes the binding constraint and sets the 206-per-category cap. The final balanced set is drawn by iterative high-confidence sampling (seed 42; medqa_focused_1030_summary.json). Inter-method agreement thus serves as a built-in label-quality control; an additional human audit and confusion matrix [TBD-EXP7] would further quantify residual category noise.

Quality control (verified). Across all 46,350 model generations, **0** produced a malformed answer (every output yields a valid A-D letter under the first-valid-letter extraction rule), so no items are dropped for extraction failure.

Example question types. Clinical Findings — interpreting a presented sign/lab toward a finding; Diagnosis — selecting the most likely disease; Mechanism — identifying the underlying pathophysiology; Next Step/Workup — choosing the next diagnostic action; Treatment/Management — selecting management. (Representative items per category to be shown.)

Limitations of exam-style questions. MedQA items are multiple-choice and exam-derived; they reward recognition among curated distractors rather than open-ended clinical reasoning, and they do not reflect real patient encounters (Section 8).

Table 1 — Dataset category distribution. Five categories $\times$ 206 = 1,030; gold-answer letter distribution A/B/C/D $\approx$ 261/278/251/240 (mild, not significant). Caption should state the balancing constraint and seed.

4. Experimental Setup

Purpose: make the evaluation fully reproducible.

Models. Five open **base** models: Qwen2.5 7B / 14B / 32B and OLMo-3 7B / 32B (HF IDs in configs/). All are pretrained, non-instruction-tuned, isolating language-model behavior from tuning effects.

Serving and environment. vLLM 0.14.1, completions endpoint (use_chat_api: false), bfloat16, tensor-parallel 2, GPU-memory utilization 0.7, max model length 4096, max_output_tokens 800, on NVIDIA A100-SXM4-80GB; checkpoint every 10 questions.

Prompt format. A simple zero-shot prompt presenting the question and options with no “you are a medical expert” preamble and no few-shot exemplars. We [TBD-EXP3: report a prompt-format robustness ablation] to show findings are not prompt artifacts.

Decoding and runs. Temperatures 0.0, 0.3, 0.7; **3 independent runs** per (model, temperature) — 45 runs, 46,350 question-evaluations total.

Answer extraction. The predicted answer is the **first valid A-D letter** in the generation. We note a base-model behavior relevant to extraction: Qwen2.5 base models do not reliably emit a stop token and run to the 800-token cap on $\sim$ 75% of items at

$t=0.0$ (often generating a new question after answering), whereas OLMo-3 base models stop almost immediately (≈ 4 tokens for 7B). Because the answer letter appears first, extraction is reliable despite this verbosity (Section 6, Table on token behavior).

Accuracy and malformed outputs. Accuracy = fraction with extracted letter equal to gold; no outputs were malformed.

Statistical methods. Wilson score 95% CIs for accuracy (overall $n=1,030$; per category $n=206$); exact McNemar tests for paired model comparisons; paired bootstrap (10k resamples) for effect-size CIs; Holm correction across comparison families; a logistic-regression likelihood-ratio test for category $\times$ temperature interaction; chi-square for letter-bias control.

Reproducibility artifacts. Configs (configs/), runner (scripts/base_runs/), SLURM jobs (slurm/full_run/), raw per-question results (results/base_runs/), and analysis scripts (papers/paper1/analysis/) are released. **Table 2** lists models and settings.

5. Results

Purpose: report the verified empirical findings with uncertainty.

5.1 Overall accuracy (Table 3, Figure 2). At greedy decoding ($t=0.0$, mean of 3 runs): OLMo-3 7B **46.18%**, OLMo-3 32B **58.64%**, Qwen2.5 7B **67.96%**, Qwen2.5 14B **68.48%**, Qwen2.5 32B **73.40%** — all far above the 25% random baseline. Greedy decoding is essentially deterministic (run-to-run std ≤ 0.05 pp).

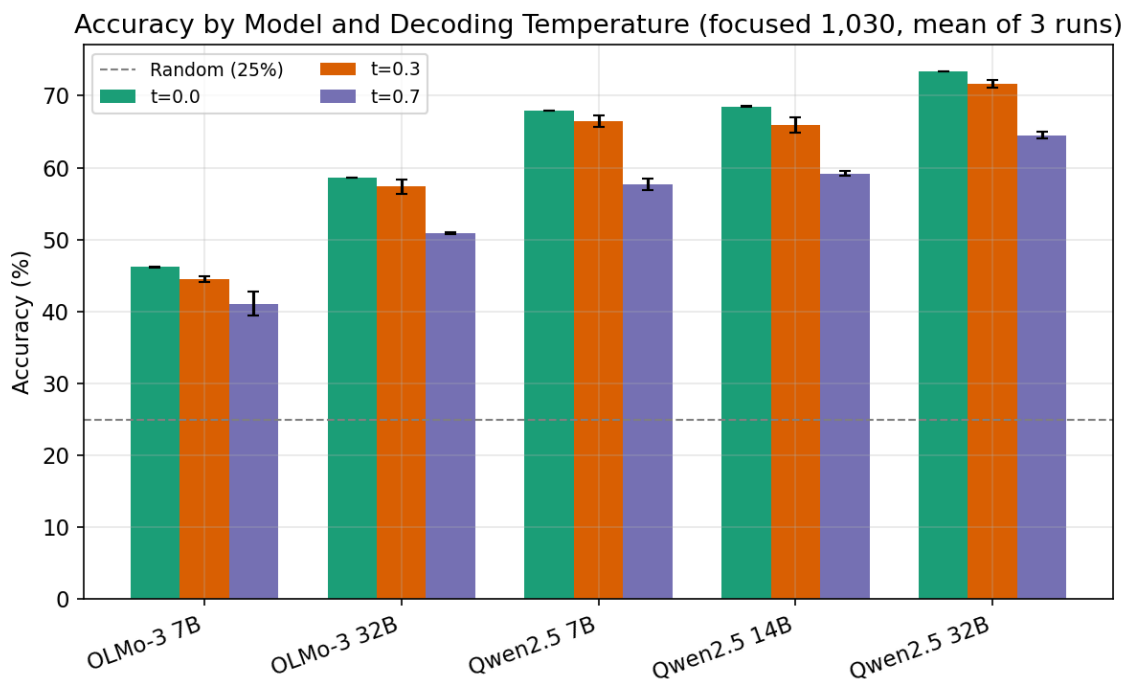


Figure 2. Accuracy by model and decoding temperature (mean of 3 runs; error bars = run std).

5.2 Category-resolved accuracy (Table 4, Figures 3 and 6). Clinical Findings is the hardest category for every model; Treatment/Management and Mechanism/Pathophysiology are consistently easiest. Per-category Wilson 95% CIs (n=206) confirm the ordering is not noise (Figure 6). This structure is invisible in the aggregate.

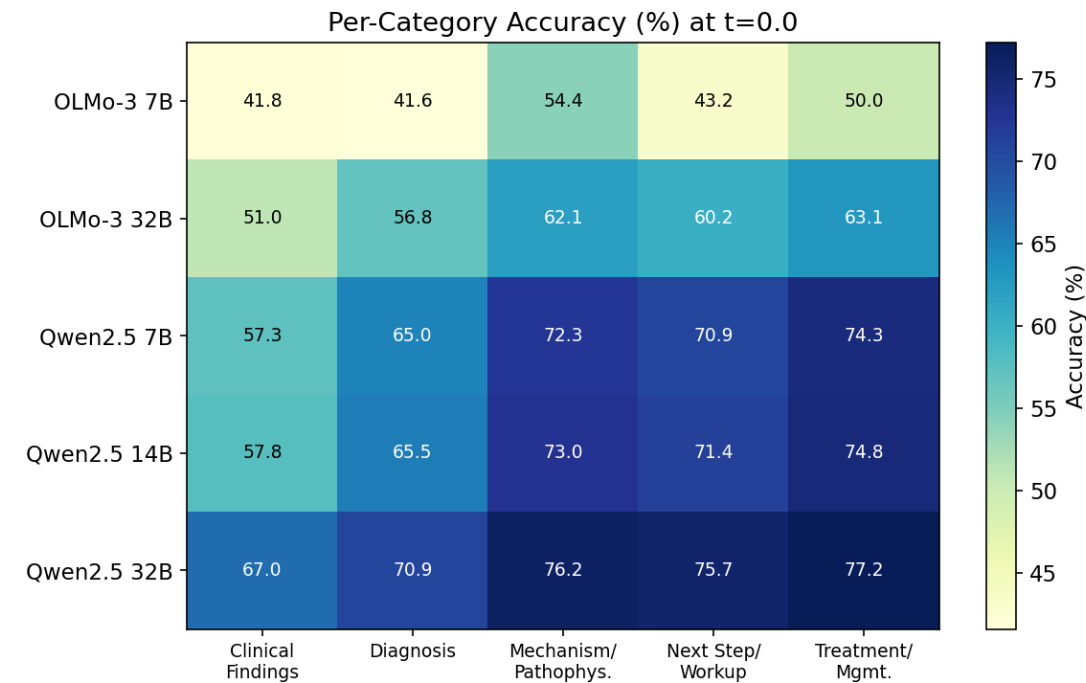


Figure 3. Per-category accuracy heatmap at t=0.0.

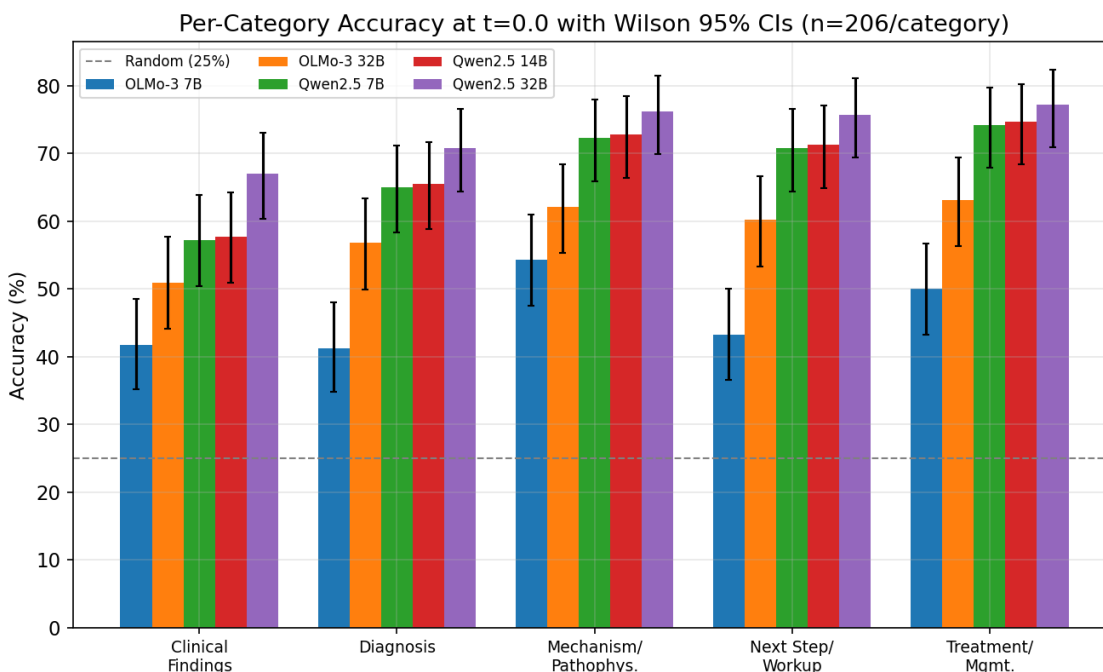


Figure 6. Per-category accuracy at $t=0.0$ with Wilson 95% CIs ($n=206/\text{category}$).

5.3 Temperature sensitivity (Table 7). Higher temperature **monotonically degrades** accuracy for all models (overall $t=0.0 \rightarrow t=0.7$ drop $\approx 3\text{--}10$ pp; bootstrap CIs in sectionB_temp_drop_ci.csv). Greedy decoding is best across the board.

5.4 Answer stability (Table 5, Figure 4). Run-to-run **answer instability** — the share of questions whose predicted letter is not identical across the 3 runs — rises from $\approx 0\%$ at $t=0.0$ to **44.66% (Qwen2.5 32B) - 67.57% (OLMo-3 7B)** at $t=0.7$. Stronger/larger models are more stable at equal temperature.

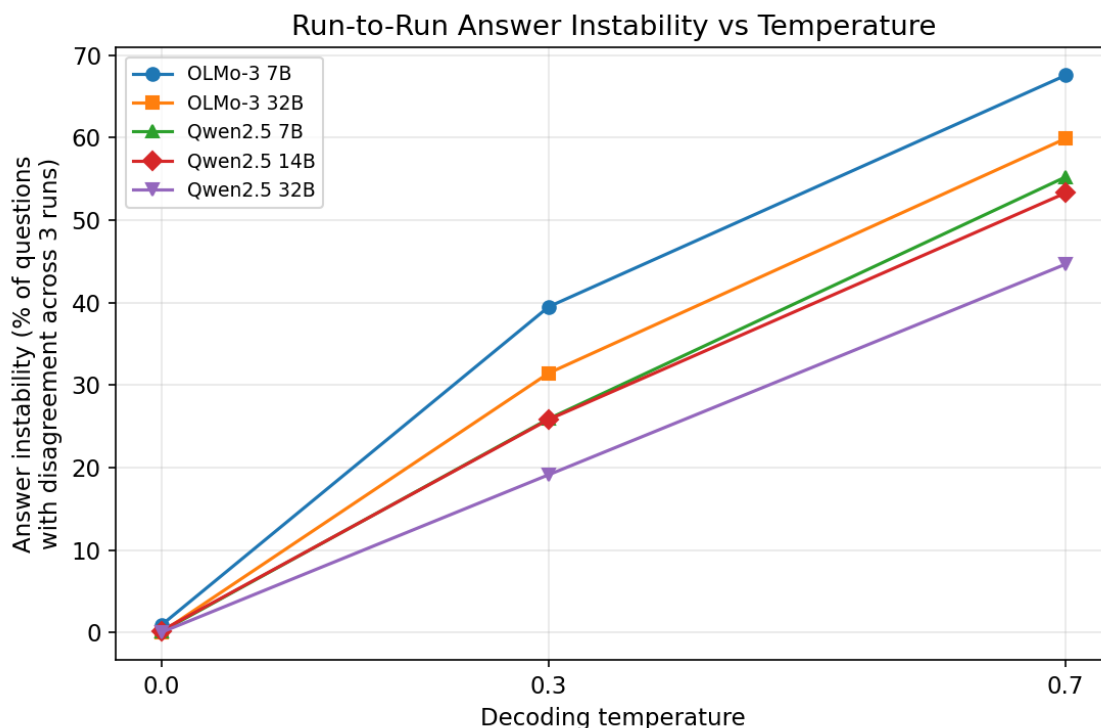


Figure 4. Run-to-run answer instability vs decoding temperature.

5.5 Self-consistency (Table 6). Majority vote over 3 samples at $t=0.7$ recovers **+3.4 to +4.2 pp** but **never reaches greedy accuracy**; net per-item effect of sampling vs greedy is negative for every model. [TBD-EXP5: self-consistency curve for $k=5/10/20$.]

5.6 Scale and family (Tables 3, 8; McNemar/bootstrap). Qwen2.5 dominates OLMo-3 at every scale. Two statistically grounded results: the **Qwen2.5 7B→14B difference is not significant** ($\Delta+0.49$ pp, bootstrap CI $[-0.10, 1.07]$, McNemar $p=0.18$, survives Holm); and **Qwen2.5 7B significantly outperforms the 4× larger OLMo-3 32B** ($\Delta+9.32$ pp, $p\approx 6e-9$).

5.7 Letter-bias control. No model’s predicted-letter distribution deviates significantly from gold (all $p>0.11$): performance is not a positional artifact.

5.8 Cross-model error structure. Of 1,030 questions, **328 (32%)** are solved by all five models and **144 (14%)** are missed by all five; this “hard core” skews toward Clinical Findings (41/144). [TBD-EXP6: expert review of the hard core: error types and ambiguity rate.]

6. Analysis

Purpose: interpret the patterns and test interactions.

6.1 Scale effects are uneven. Within OLMo-3, 7B→32B adds +12.5 pp; within Qwen2.5, 14B→32B adds +4.95 pp (significant) but 7B→14B is negligible and non-

significant. Across families, pretraining quality dominates raw scale (Qwen2.5 7B > OLMo-3 32B). The practical reading: parameter count is a poor predictor of medical-QA accuracy across model families.

6.2 Temperature acts as two main effects, not an interaction. Although per-category drops differ descriptively, a logistic-regression LR test finds **no significant category×temperature interaction** for 4 of 5 models (significant only for Qwen2.5 14B, $p=0.02$). We therefore report category difficulty and temperature degradation as independent main effects: every category degrades, roughly proportionally, as temperature rises.

6.3 Stability as a confidence signal. At $t=0.7$, questions on which all three runs agree (“stable”) are far more often correct than unstable ones — a **+22.9 to +42.2 pp** accuracy gap. Cross-run stability is thus a logit-free confidence proxy, and a logprob re-run (EXP1) confirms it: per-question greedy confidence (softmax over the four option-letter logprobs) correlates **negatively with $t=0.7$ answer instability for every model** ($r = -0.34$ to -0.62) — higher confidence, lower instability. Models are reasonably calibrated (ECE 0.024–0.078; reliability diagram in Figure 7), so the stability signal reflects genuine model confidence, not an artifact. This is the empirical hook for the verification/abstention methods in later work.

Reproducibility note (EXP1). The logprob re-run repeated greedy decoding three times per model. OLMo-3 7B/32B and Qwen2.5 7B are bit-for-bit deterministic across launches, but the two larger Qwen models are mildly **nondeterministic** even at $t=0.0$ (Qwen2.5 14B / 32B agree on 97.2% / 98.2% of questions across runs; ≈ 0.29 pp accuracy spread), consistent with non-deterministic GPU kernels / tensor-parallel reductions — hence we report run medians. One caveat: Qwen2.5 7B reproduced 59.71% in the logprob environment vs 67.96% in the main benchmark environment (a reproducible 8.3 pp gap traced to a differing model snapshot / vLLM runtime, not to decoding); calibration for Qwen2.5 7B is reported from the logprob environment.

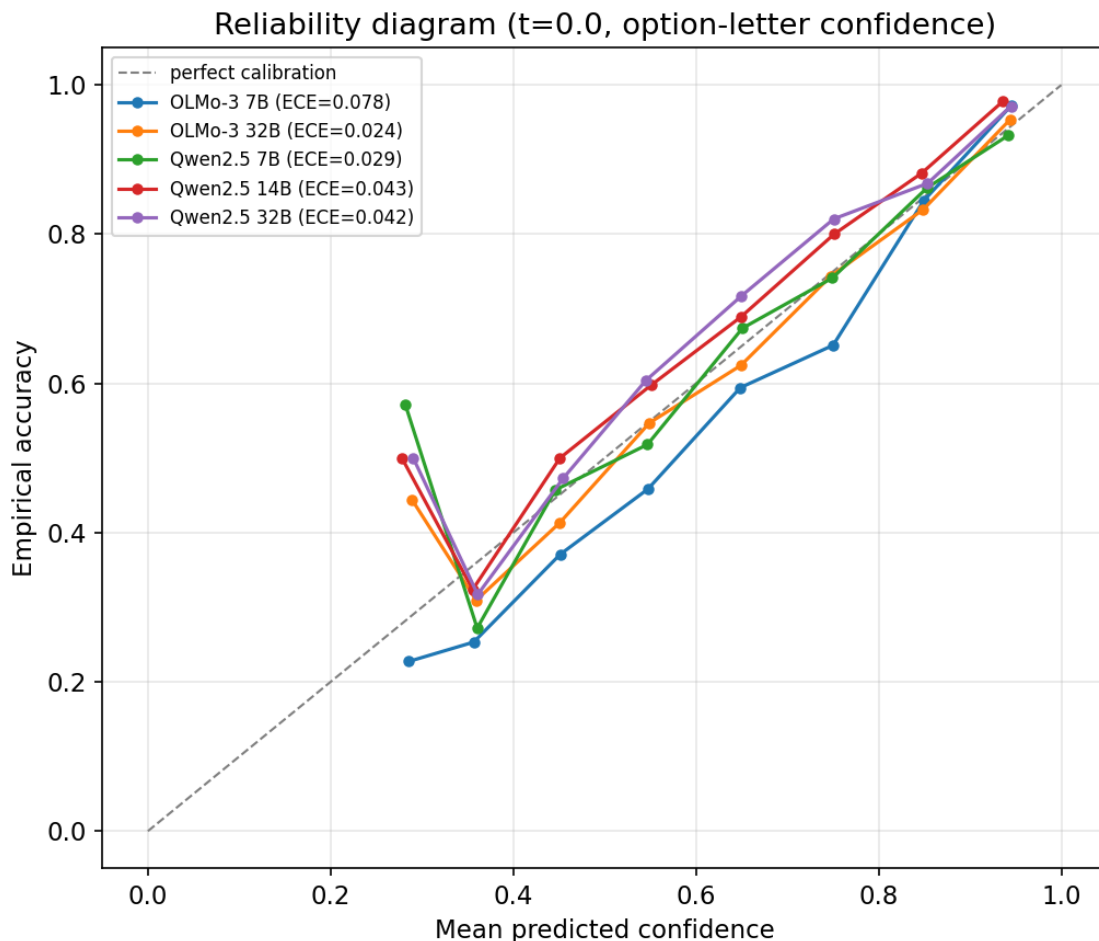


Figure 7. Reliability diagram ($t=0.0$, option-letter confidence): predicted confidence vs empirical accuracy, per model (EXP1).

6.4 When sampling helps vs hurts. Sampling rescues a non-trivial set of items wrong under greedy (32–90 per model) but loses more (80–124), so greedy dominates on average while not dominating item-by-item — consistent with stochastic exploration occasionally escaping a confident wrong mode. [TBD-EXP5: does self-consistency at higher k change this balance?]

6.5 Benchmark construction changes the scaling story. On the full 12,723-question 5-option set, Qwen2.5 7B→14B gains +7.18 pp; on the focused balanced 4-option set the same step is **+0.52 pp**. Aggregate, unbalanced evaluation and balanced evaluation can yield *different scaling conclusions* — a methodological caution central to the paper’s thesis (reconciliation table in RESULTS.md).

6.6 Generation behavior and extraction. The Qwen-vs-OLMo stopping contrast ($\approx 75\%$ cap-hit vs ≈ 4 tokens) justifies the first-valid-letter rule and explains why greedy results are reliable despite base-model verbosity; wrong answers are associated with slightly longer generations across models.

6.7 Error analysis framework. We organize wrong answers using the taxonomy in BRIEF.md §13 (misread finding, incorrect diagnosis, mechanism confusion, treatment-

guideline confusion, next-step sequencing, distractor attraction, overgeneralization, ignored age/sex/risk, temporal-reasoning, pharmacology confusion, extraction failure — the last empirically ~ 0). [TBD-EXP6: distribution of error types over the hard core, with examples



Figure 5. Error rate by reasoning category ($t=0.0$, run 1).

from `error_samples.md`.]

7. Discussion

Aggregate MedQA accuracy is a coarse instrument: it conceals a stable category-difficulty ordering, is sensitive to decoding choices that are rarely reported, and can invert scaling conclusions relative to a balanced evaluation. Reporting category-resolved accuracy with uncertainty, greedy-vs-sampled behavior, and run-to-run stability gives a more faithful picture of what base models have and have not learned about medical reasoning. The stability signal in particular suggests that *how consistently* a model answers carries information about *whether* it is right — a property we exploit in later work on verification and neuro-symbolic grounding. We frame diagnosis there as coarse-to-fine evidence acquisition rather than flat answer selection; the present benchmark supplies the empirical baseline and the failure modes that motivate that program (Section 9). We make no claims about clinical deployment or patient safety.

8. Limitations

MedQA is exam-style and multiple-choice; high accuracy reflects recognition among curated distractors, not open-ended clinical reasoning or real patient care. Base models are not the instruction-tuned assistants people actually use, so results characterize raw language models rather than deployed systems [TBD-EXP2: base-vs-instruct contrast]. The reasoning categories are introduced in this work (MedQA provides none) and embed classification assumptions; they are assigned by dual-method (rule + LLM) agreement, which controls for label noise but does not eliminate it [TBD-EXP7: human agreement audit]. Accuracy does not capture reasoning quality or explanation faithfulness. Temperature effects may depend on prompt format and the answer-extraction rule [TBD-EXP3: prompt ablation], though the first-valid-letter rule was empirically robust (0 malformed outputs). All evaluation is on a single benchmark and five open models; generalization to other datasets and proprietary models is untested. We make no patient-safety claims.

9. Conclusion

We presented a category-balanced, temperature-controlled, statistically grounded benchmark of five open base language models on USMLE-style medical questions. Aggregate accuracy hides a consistent category-difficulty ordering (Clinical Findings hardest); decoding temperature monotonically degrades accuracy while sharply increasing run-to-run answer instability; scale helps unevenly, with a smaller Qwen model significantly beating a 4× larger OLMo model and a non-significant Qwen 7B→14B step on the balanced set; and cross-run stability predicts correctness. These results argue for category-resolved, stability-aware medical-reasoning evaluation and establish the empirical baseline and failure modes that motivate our subsequent work on UMLS-grounded verification, chain-of-verification, contradiction detection, and agentic differential diagnosis. Data, configurations, and analysis code are released for full reproducibility.

Appendix / Reproducibility (outline)

- A. Exact prompt template(s) and chat templates [TBD-EXP2/EXP3 variants].
- B. Full per-model, per-temperature, per-run tables (from tables/).
- C. Statistical method details (Wilson, McNemar, bootstrap, Holm, LR interaction test).
- D. Hardware/software manifest and SLURM scripts.
- E. Hard-core question list and error samples (error_samples.md).
- F. Full-set vs focused-set reconciliation.

Figures and tables (asset map)

- Table 1 dataset distribution · Table 2 models/settings · Table 3 overall accuracy · Table 4 category accuracy · Table 5 variability/instability · Table 6 majority vote

- Table 7 temperature sensitivity · Table 8 scale · Table 9 error distribution.
- Figure 1 study-design overview (*to draw*) · Figure 2 accuracy by model×temp (figures/fig2_...png) · Figure 3 category heatmap (fig3_...) · Figure 4 answer instability (fig4_...) · Figure 5 error by category (fig5_...) · Figure 6 per-category accuracy w/ CIs (fig6_...) · Figure 7 reliability diagram (fig7_reliability.png, EXP1) · [TBD-EXP5: self-consistency curve].

Publication strategy (notes)

arXiv-first. Target ML/medical-AI workshops (e.g., ML4H, clinical-NLP venues) for the current version; for a full conference/journal submission add EXP1 (calibration) and EXP2 (instruct comparison), which address the two most likely reviewer objections. Biomedical informatics venues (e.g., AMIA) are viable with EXP6/EXP7 dataset-validation additions.